MAX 507 / MSC 655 – Visual Analytics – Data Analytics and Visualization

Spring 2026

Data Visualization Project – Proposal (Group 4)

AI in Marketing: When Efficiency Gains and Channel Disruption Happen at the Same Time

A Data-Driven Framework for Reallocating Marketing Spend

Background and Motivation

We chose this topic because of something we noticed while reading industry reports on AI and marketing: everyone talks about these two trends separately, but nobody puts them together.

The first trend is that AI tools are making marketing work faster and cheaper. According to the Gartner 2025 CMO Spend Survey, 49% of Chief Marketing Officers say AI has improved their time efficiency, and 40% say it has reduced costs. Marketing teams are using AI to write blog posts, generate ad copy, personalize emails, and automate campaign workflows. Budgets are tight, marketing spending has been flat at 7.7% of company revenue for two years. CMOs are cutting agency contracts (39%) and headcount (39%) to fund AI tools instead (Gartner, 2025).

The second trend is that AI is simultaneously changing how people find things online. Google now shows AI-generated summaries (called 'AI Overviews') at the top of search results, which answer questions directly without requiring users to click through to websites. A SEMrush study of over 10 million keywords found that these AI Overviews appeared on 6.49% of searches in January 2025, peaked at nearly 25% in July, and settled at 15.69% by November. That means for roughly 1 in 6 searches, Google is answering the question itself. For marketing teams that rely on organic search traffic to reach customers, this is a problem.

What caught my attention is that these two trends are on a collision course. Companies are using AI to produce more content, faster and cheaper, and then pushing that content into channels (like organic search and blog content) that are losing effectiveness because of AI. A CMO might celebrate a 40% reduction in content production costs while not realizing that the channel delivering that content to customers has lost a third of its click-through rate. The savings on one side could be wiped out by losses on the other.

But here is the important nuance: not all channels are affected equally. Email marketing, paid social media, and retail media (ads on Amazon, Walmart, etc.) do not operate through Google search at all. They are completely unaffected by AI Overviews. So, the question is not just 'is AI good or bad for marketing?', it is 'which specific channels benefit from AI tools without being hurt by AI-driven search disruption, and which ones are getting hit from both sides?'

I could not find any existing dashboard, report, or student project that combines these two forces into a single view. That is the gap this project fills.

What we want to Learn and Accomplish

The goal of this project is to build a visual framework in Tableau that helps marketing leaders decide where to invest their budgets by looking at three things together:

- Where is advertising money actually flowing? (market momentum)
- Which channels are being disrupted by AI search features? (discovery risk)

- Where are AI efficiency gains real vs. where are they being absorbed by declining channel effectiveness? (efficiency-to-outcome gap)

The output is a simple four-category recommendation for each channel: Scale (increase investment), Test (experiment cautiously), Hold (maintain current levels), or Reduce (pull back). We also want to build an interactive scenario tool so users can test 'what if' questions – like 'what happens to my allocation if AI Overviews expand to 30% of searches?'

Intended Audience

The primary audience is marketing leaders – CMOs, VP Marketing, and anyone responsible for deciding how to split a marketing budget across channels. These are people who need to make allocation decisions now, with imperfect information, in a market that is changing quickly. The secondary audience includes CFOs evaluating whether AI marketing investments are actually paying off, and digital strategy consultants advising companies on where to spend.

Five Questions

Question 1: Which advertising channels are growing and which are stalling?

U.S. digital advertising hit $258.6 billion in 2024 (IAB/PwC). But the growth is very uneven: social media advertising grew 36.7% year-over-year, retail media grew 23%, while search, still the largest channel at $102.9 billion, grew 15.9% but has been slowly losing market share (from 42.2% in 2020 to 39.8% in 2024). We want to map these trajectories over 5 years to see which channels have sustained momentum and which are plateauing. This matters because a channel that is losing share may not be the right place to increase AI-powered content production.

Question 2: Is there a gap between AI efficiency gains and actual business results?

The Gartner and CMO Survey data show a striking disconnect: 49% of CMOs say AI saves time, but only 27% say those savings translate into better business outcomes. That is a 22-percentage-point gap. We want to understand whether this gap is the same across industries and company sizes, or whether certain segments are converting efficiency into results while others are not. Even with a small survey (n=281), directional patterns across industries would be more useful than the overall average.

Question 3: Which channels are most exposed to AI-driven search disruption, and which are insulated?

The SEMrush data shows something we did not expect: AI Overviews started out targeting informational queries (91.3% in January), but by October, informational queries had dropped to 57.1% while commercial and navigational queries surged. Navigational queries with AI Overviews went from 0.74% to 10.33%, a 14x increase. This means AI is no longer just affecting blog posts and how-to guides; it is moving into product comparisons and branded searches. We want to map exactly which channels depend on which query types, because that determines their exposure.

Question 4: When you combine all three factors – market momentum, AI efficiency, and discovery risk, which channels come out positive and which come out negative?

This is the central question. We plan to combine the three factors into a single score I am calling the 'Net Allocation Pressure Score' (NAPS). The idea is simple: channels that have strong market momentum AND benefit from AI efficiency AND face low search disruption should be candidates for more investment. Channels where the disruption outweighs the efficiency gains should be candidates for pullback. We have already run the baseline calculations, and the results make intuitive sense: Paid Social and Organic Social classify as 'Scale' (high momentum, zero search exposure), while Organic Search classifies as 'Reduce' (moderate momentum but highest disruption exposure). Content/Blog lands in

'Hold', it has the highest AI efficiency gains of any channel, but those gains are almost exactly offset by its high exposure to AI Overviews.

Question 5: How do these recommendations change under different assumptions about the future?

Any framework based on current data is just a snapshot. What if AI Overviews expand to cover 30% or 40% of searches? What if AI efficiency gains plateau as the early hype fades? We want to build Tableau parameters that let the user adjust these assumptions and watch the allocation recommendations shift in real time. For example, when we increase the AI disruption multiplier to 2x in my scenario model, Content/Blog drops from 'Hold' to 'Reduce,' and Organic Search drops deep into 'Reduce' territory. This kind of sensitivity testing turns the dashboard from a static report into a tool people can actually use.

Intent of the Project

This is an explanatory project. It has a thesis: AI efficiency gains do not automatically translate into marketing growth because the channels where that efficiency is being deployed are simultaneously losing effectiveness due to AI-driven search disruption. The right allocation strategy is channel-specific, some channels benefit from AI on both sides, while others are caught in a squeeze. The project explains this interaction and provides a framework for acting on it.

Datasets

This project uses two primary public datasets, one supplementary reference dataset, and one generated scenario model. All public datasets have been extracted and are ready to load into Tableau. Source URLs are provided for each.

Dataset 1: IAB/PwC Internet Advertising Revenue (2020–2024)

Source: Interactive Advertising Bureau (IAB) and PricewaterhouseCoopers (PwC). These are the standard industry reports for U.S. digital advertising revenue. Published annually as free PDF downloads at iab.com/insights/internet-advertising-revenue-report-2024/ (2024 report), with prior years at iab.com/topics/ad-revenue/.

Format: We extracted the key revenue-by-channel tables from each of the 5 annual PDF reports and consolidated them into a single CSV file with 30 rows (6 channels x 5 years).

The columns in the extracted dataset:

Column	Type	Description
Year	Date (2020–2024)	Reporting year
Channel	Categorical	Search, Social Media, Display, Digital Video, Retail Media, Audio/Podcast
Revenue_Billions	Continuous ($B)	Total U.S. revenue for that channel in that year
YoY_Growth_Pct	Continuous (%)	Year-over-year growth rate
Share_of_Total_Pct	Continuous (%)	Channel's share of the total digital ad market
Total_Market_Billions	Continuous ($B)	Total U.S. digital ad market size that year
Source	Text	Specific IAB report referenced

Geographic location: United States (national aggregate).

How it helps: This dataset directly answers Question 1. When we plotted the 5-year trajectories (see Visualization 1 below), the story became clear: Social media went from $41.5B to $88.8B in five years. Search is still the biggest channel at $102.9B but its share has been slowly declining (42.2% in 2020 to 39.8% in 2024). Digital video nearly tripled. These trajectories feed into the Market Momentum Index (MMI), a score we compute for each channel based on its growth rate and share trend. Channels with high and accelerating growth get high MMI scores.

Limitations:

- National aggregate only – no breakdown by company size or industry.
- Shows where money is flowing, not whether it is working. Revenue does not equal ROI.
- IAB reclassified some categories over the 5 years (e.g., 'Retail Media' was not tracked separately before 2022), so we had to harmonize channel names across years.

Data cleaning:

- Extracted tables from 5 PDF reports using Tabula and manual cross-referencing. Standardized channel names across years (created a mapping table for categories that changed). Calculated derived fields: 5-year CAGR, share shift, and MMI score. Reconciled minor rounding discrepancies between IAB summary tables and detail tables.

Dataset 2: SEMrush AI Overviews Impact Study (January–November 2025)

Source: SEMrush, one of the largest SEO analytics platforms. They published a study analyzing 10 million+ keywords to track how Google's AI Overviews are changing search behavior. Available at semrush.com/blog/semrush-ai-overviews-study/.

Format: The data is published as tables and charts within a blog post, not as a downloadable file. We manually transcribed the data into two CSV files:

File A, Monthly trend (11 rows): One row per month from January through November 2025.

Column	Type	Description
Month	Date (YYYY-MM)	January 2025 through November 2025
AIO_Prevalence_Pct	Continuous (%)	% of keywords showing an AI Overview
Informational_Intent_Pct	Continuous (%)	% of AI Overview queries that are informational
Commercial_Intent_Pct	Continuous (%)	% that are commercial intent
Transactional_Intent_Pct	Continuous (%)	% that are transactional
Navigational_Intent_Pct	Continuous (%)	% that are navigational
Source	Text	Notes whether the data point was directly reported or interpolated

We want to be transparent about the data: SEMrush published exact figures for January (6.49%), March, July (24.61%), October, and November (15.69%). For months where they showed charts but did not publish exact numbers, we interpolated from the visible trend lines. Every interpolated value is flagged in the Source column.

File B, Industry saturation (19 rows): AI Overview penetration by industry vertical as of November 2025. Ranges from Science (25.96%) at the top to Real Estate (2.00%) at the bottom.

Geographic location: Global aggregate with heavy U.S./EU focus.

How it helps: This dataset directly answers Question 3. The most important finding and the one that surprised me is the intent migration. In January, 91.3% of AI Overview queries were informational (blog-type questions). By October, that dropped to 57.1%, while commercial queries (product comparisons, buying guides) surged to 18.57% and navigational queries (branded searches) went from 0.74% to 10.33%. This means AI Overviews are no longer just affecting informational blog content, they are expanding into the kinds of searches that directly drive revenue. This intent migration is central to the AI Discovery Exposure Index (ADEI) we compute for each channel.

Limitations:

- Published as a blog post, not a downloadable dataset. Required manual transcription.
- Aggregated at monthly level, no individual keyword data available.
- 11 months of data is useful for directional trends but too short for time-series forecasting.
- Only covers Google AI Overviews, does not capture ChatGPT or Perplexity search behavior.

Data cleaning:

- Manually transcribed tables from the blog post. Cross-referenced chart visuals against text-reported figures to catch rounding differences. Standardized month format to YYYY-MM. Calculated month-over-month change rates and ADEI scores per industry.

Dataset 3: Gartner CMO Spend Survey 2025 + CMO Survey Spring 2025 + FirstPageSage CAC Benchmarks

This combines three publicly available sources into a unified AI efficiency and cost reference:

- Gartner 2025 CMO Spend Survey: press release with headline metrics on marketing budgets and AI productivity. Source: gartner.com (published May 12, 2025).
- CMO Survey, Spring 2025 (Duke University / Deloitte): survey of 281 U.S. marketing leaders (99% VP-level or above) conducted January–February 2025. Source: cmosurvey.org/results/spring-2025/.

- FirstPageSage CAC by Channel 2026 Benchmarks: Customer Acquisition Cost benchmarks for 22 marketing channels (B2B and B2C), based on ~120 firms. Source: firstpagesage.com/marketing/cac-by-channel-fc/.

Format: We extracted these into three small CSV files. The Gartner file has 14 rows of key metrics. The CMO Survey file has 24 rows covering AI adoption rates, outcomes, and industry distribution. The FirstPageSage file has 22 rows with B2B and B2C CAC per channel.

Key dimensions: Industry sector, company size, metric category (budget, AI productivity, workforce). Key measures: Marketing budget as % of revenue (7.7%), AI time efficiency (49%), AI cost efficiency (40%), business outcome translation (27%), agency spend cuts (39%), labor cost cuts (39%), CAC by channel in USD.

Geographic location: United States.

How it helps: This answers Question 2, the efficiency-to-outcome gap. The 22-point gap between '49% say AI saves time' and '27% say it produces results' is the core insight. The FirstPageSage CAC data grounds the analysis in real dollar costs. For example, B2B basic SEO costs $1,786 per customer acquisition while email costs only $510, which aligns with the NAPS model's recommendation to favor email over organic search.

Limitations:

- Gartner data is from a press release summary, not a full dataset.
- CMO Survey is self-reported (CMOs may overstate AI benefits) and small (n=281). Industry cross-tabs will have small cell sizes.
- FirstPageSage data covers ~120 firms and skews toward higher-value B2C industries.

Dataset 4 (Generated): Allocation Pressure Scenario Model

This dataset is what ties everything together. It is not simulated company data or made-up numbers. It is a structured combination of the three scores we derived from the public datasets, crossed with dimensions that allow segmented analysis.

We wrote a Python script (using pandas and numpy) that does the following:

1. Takes each of the 8 marketing channels and assigns it three scores based on the public data: a Market Momentum Index (MMI) from the IAB revenue trends, an AI Efficiency Leverage Index (AELI) from the Gartner/CMO Survey metrics, and an AI Discovery Exposure Index (ADEI) from the SEMrush data.
2. Combines these into a Net Allocation Pressure Score: NAPS = (MMI + AELI) – ADEI. Normalizes the result to a 0–100 scale and classifies each channel as Scale (75+), Test (55–74), Hold (35–54), or Reduce (below 35).
3. Crosses every channel with 6 industries, 3 company sizes (SMB, Mid-Market, Enterprise), and 10 scenario variants (Baseline, High/Low AI Disruption, High/Low AI Efficiency, Bull/Bear Market, Best Case, Worst Case).

Format: CSV with 1,440 rows (8 channels x 6 industries x 3 sizes x 10 scenarios) and 21 columns.

Key columns:

Column	Type	Description
Channel	Categorical	8 channels: Organic Search, Paid Search, Paid Social, etc.
Industry	Categorical	Tech/Software, Healthcare, Financial Services, Retail, etc.
Company_Size	Categorical	SMB, Mid-Market, Enterprise
Scenario	Categorical	Baseline, High_AI_Disruption, Low_AI_Disruption, etc.
MMI_Score	Continuous (0–100)	Market Momentum Index
AELI_Score	Continuous (0–100)	AI Efficiency Leverage Index
ADEI_Score	Continuous (0–100)	AI Discovery Exposure Index

NAPS_Score	Continuous (0–100)	Net Allocation Pressure Score
Allocation_Classification	Categorical	Scale, Test, Hold, or Reduce

How it helps: This is the engine behind Dashboard 3 and directly answers Questions 4 and 5. Under baseline conditions, the model classifies Paid Social and Organic Social as 'Scale,' Digital Video, Retail Media, and Email as 'Test,' Content/Blog as 'Hold,' and Organic Search and Paid Search as 'Reduce.' These classifications shift meaningfully across scenarios. In the Worst Case, the average NAPS drops from 56.9 to 39.9, and 'Reduce' classifications nearly double.

Important caveat: This model produces directional recommendations, not precise ROI predictions. Every weight and threshold is documented in the Python script. In the Tableau dashboards, we will label all outputs as 'the model suggests...' rather than claiming definitive answers.

Calibration:

We cross-referenced the model's output against FirstPageSage CAC benchmarks to check whether the classifications make sense. Channels classified as 'Scale' (like Email at $510 B2B CAC and Social at $658) do have lower acquisition costs than channels classified as 'Reduce' (like Basic SEO at $1,786). This alignment gives us confidence the scoring logic is directionally correct, even though the weighting choices involve judgment calls.

Visualization Plan

Type of Visualization

The project will produce 3 explanatory dashboards connected by a Tableau Story with 6 story points. The focus is explanatory. We have a specific thesis to support and a narrative to walk the audience through. The third dashboard also has an interactive component (scenario parameter sliders) that lets users test their own assumptions.

Dashboard Descriptions

Dashboard 1: Market Momentum and the AI Efficiency Gap

This dashboard answers Questions 1 and 2. The top section shows a multi-line time series of 5-year channel revenue (from the IAB data), establishing which channels are growing and which are stalling. The bottom section uses a divergence bar chart to show the efficiency-to-outcome gap from the Gartner/CMO Survey data, paired with a bar chart showing how CMOs are funding AI by cutting agencies and headcount.

Dashboard 2: Channel Exposure to AI Search Disruption

This dashboard answers Question 3. The top section shows a stacked area chart of AI Overview prevalence over 11 months, colored by query intent type, to visualize the intent migration. The bottom section has an industry exposure bar chart and a channel-by-intent heatmap showing which channels depend on which query types in search.

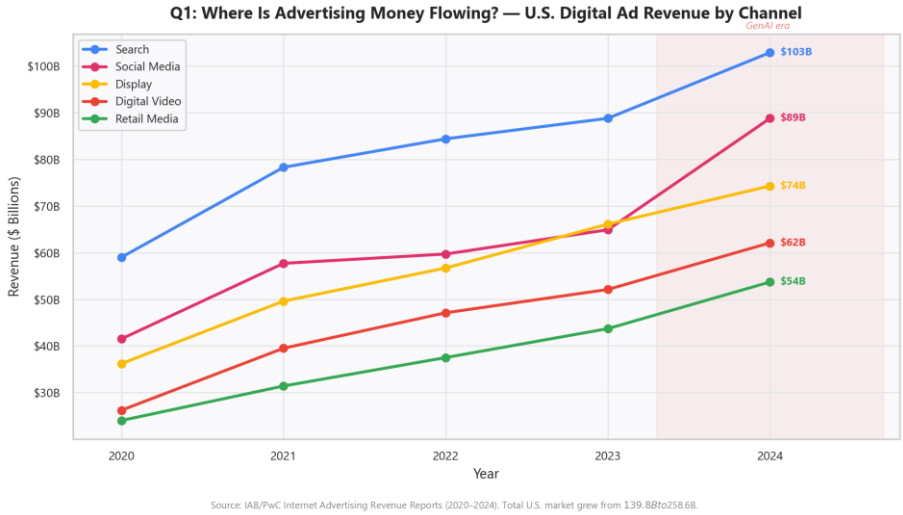
Dashboard 3: The Allocation Framework and Scenario Engine

This dashboard answers Questions 4 and 5. The main element is a quadrant scatter plot where the x-axis is positive pressure (MMI + AELI) and the y-axis is negative pressure (ADEI). Each bubble is a channel, sized by market share. Quadrant boundaries divide the space into Scale, Test, Hold, and Reduce zones. Alongside, a ranked bar chart sorts channels by NAPS score. Three Tableau parameters (sliders) let the user adjust the AI disruption level, efficiency gain assumption, and market growth rate, which updates the chart in real time.

Preliminary Visualizations

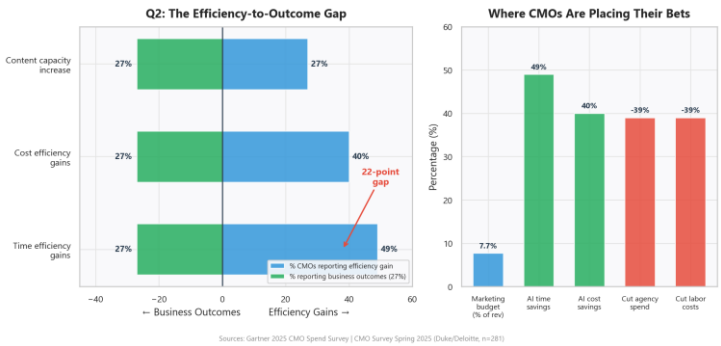
The following charts are generated from the actual project datasets using Python. They demonstrate that the data supports the questions posed above and show the direction for the final Tableau dashboards. The Tableau versions will add interactivity, filters, and story annotations.

Visualization 1, Question 1: Where is advertising money flowing?



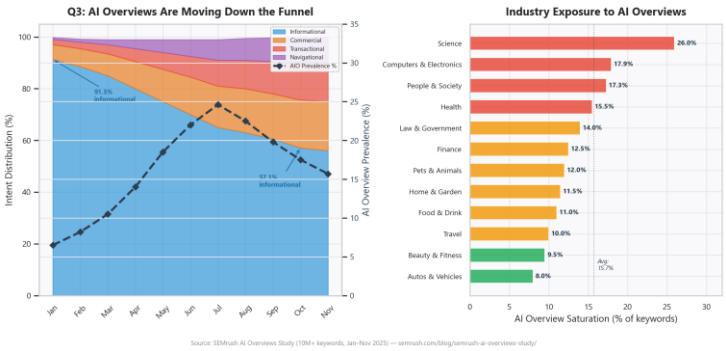
This chart uses Dataset 1 (IAB). Each line tracks a channel's revenue from 2020 to 2024. The visual makes it easy to see that Social Media has been accelerating (especially 2023–2024, where it jumped from $64.9B to $88.8B), while Search, though still the largest at $102.9B, has a flatter growth curve. Retail Media and Digital Video are growing steadily from smaller bases. In Tableau, I will add calculated fields for CAGR and share shift, which will feed the MMI score.

Visualization 2, Question 2: The efficiency-to-outcome gap



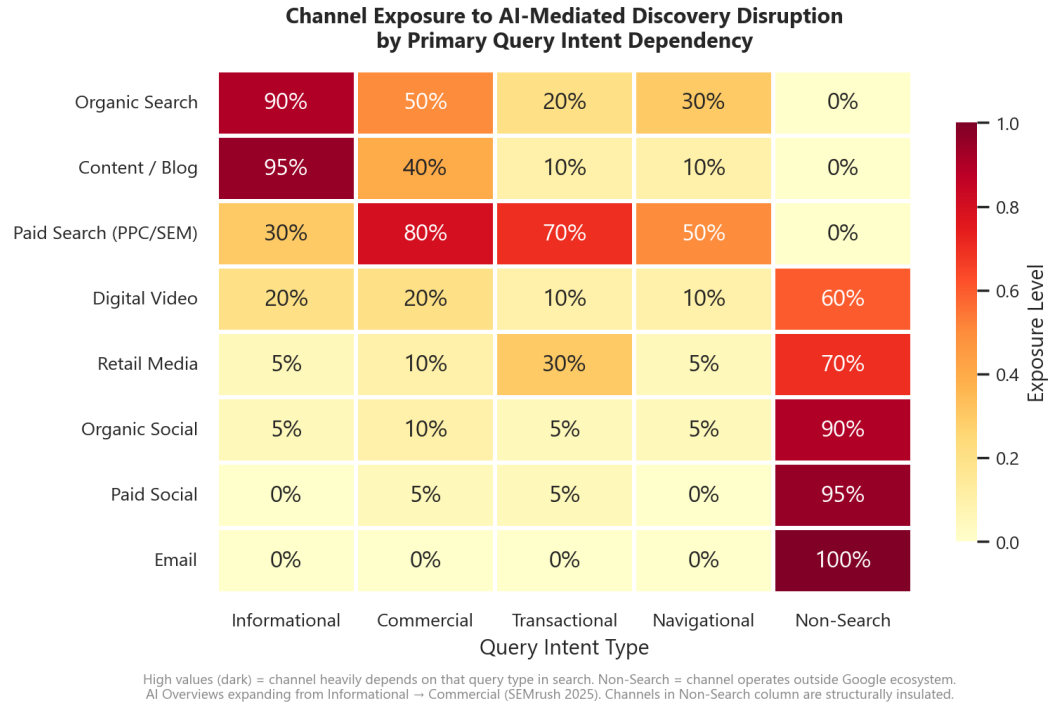
The left chart contrasts what CMOs report as AI benefits (right side: 49% time savings, 40% cost savings) against actual business outcomes (left side: 27% for all three). The 22-point gap between 'saves time' and 'produces results' is the first signal that AI marketing ROI may be overstated. The right chart shows how CMOs are funding AI: cutting agencies (39%) and cutting labor (39%) while budgets stay flat at 7.7% of revenue. In Tableau, we plan to segment this by industry if the CMO Survey cross-tabs support it.

Visualization 3a, Question 3: AI Overviews are moving down the funnel



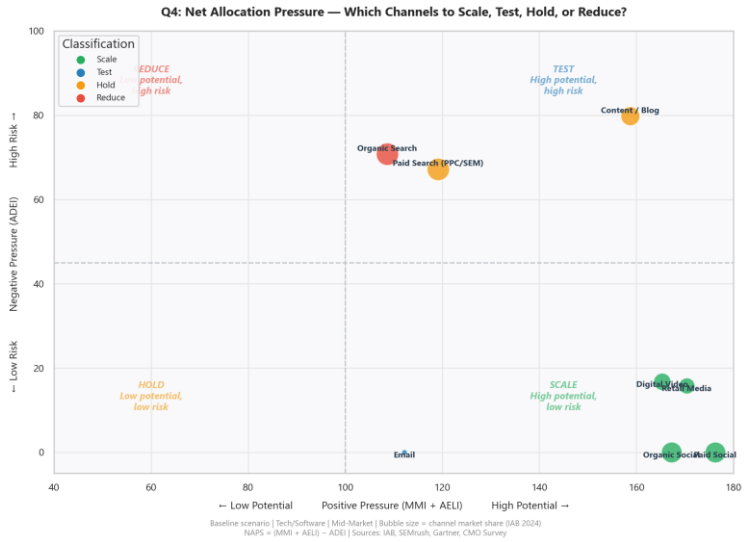
The left chart is the one that convinced us this topic was worth pursuing. The stacked area shows how the composition of AI Overview queries changed over 11 months: the blue (informational) layer shrinks from 91.3% to 57.1% while orange (commercial), red (transactional), and purple (navigational) all grow. The dashed line shows total AI Overview prevalence peaking at 24.6% in July before settling at 15.7%. The right chart ranks industries by AI Overview saturation. Science (26.0%) and Computers & Electronics (17.9%) are most exposed; Shopping (2.5%) and Real Estate (2.0%) are least exposed.

Visualization 3b, Channel exposure heatmap



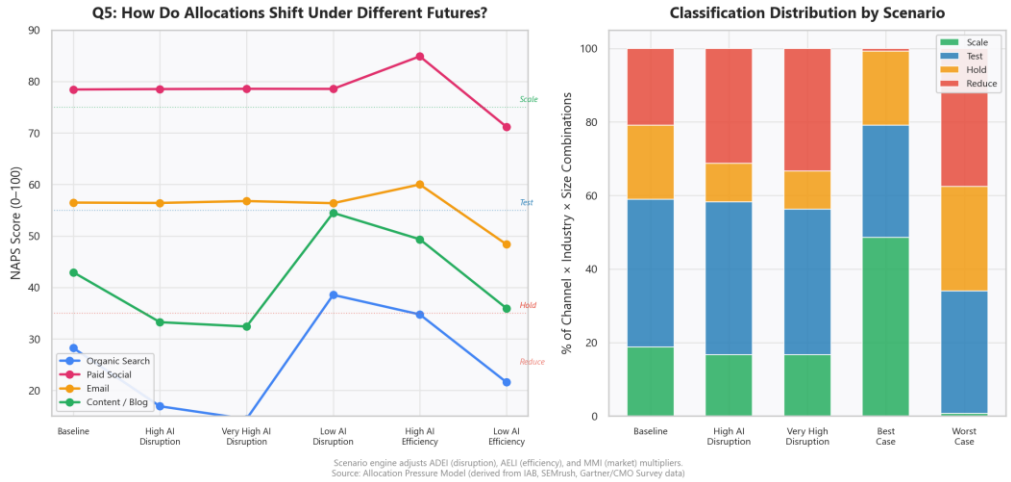
This heatmap maps each channel (rows) against query intent types (columns). The pattern tells the story clearly: channels at the top (Organic Search, Content/Blog) are dark red in the Informational and Commercial columns, they depend heavily on query types that AI Overviews are targeting. Channels at the bottom (Email, Paid Social) are entirely in the 'Non-Search' column, they do not operate through Google at all. This visual bridges Questions 3 and 4.

Visualization 4, Question 4: The allocation quadrant



This is the chart that brings everything together. Each bubble is a channel, plotted by its positive pressure (how much market momentum and AI efficiency it has, x-axis) against its negative pressure (how exposed it is to AI search disruption, y-axis). Bubble size reflects 2024 market share. The four quadrants correspond to the Scale/Test/Hold/Reduce classification. Under baseline assumptions (Tech/Software, Mid-Market), Paid Social and Organic Social land in 'Scale' (high potential, low risk), while Organic Search and Paid Search land near the 'Reduce' zone (moderate potential but high risk). Content/Blog is in an interesting position. Very high on both axes, meaning it has huge AI efficiency gains but equally huge disruption exposure, netting out to 'Hold.' In Tableau, the quadrant boundaries will shift when users adjust scenario parameters.

Visualization 5, Question 5: How recommendations shift under different scenarios



The left chart tracks four channels across six scenarios. The key takeaway: Paid Social stays in the 'Scale' zone no matter what, it is structurally resilient. Organic Search drops from borderline 'Reduce' to deep 'Reduce' as AI disruption increases. Content/Blog is the most sensitive channel, it swings between 'Hold' and 'Reduce' depending on assumptions. Email stays relatively stable in the 'Test' zone. The right chart shows the overall distribution: in the Worst Case scenario, 'Reduce' classifications dominate (roughly 40% of all combinations), while in the Best Case, 'Scale' and 'Test' expand significantly. In Tableau, this will be interactive. Users drag sliders to see these shifts happen live.

Tableau Story: Narrative Arc

The Tableau Story will have 6 story points that build a narrative from context to action:

#	Story Point	What It Shows	Narrative Purpose
1	The Starting Point	Headline metrics: budgets flat at 7.7%, 49% report AI efficiency gains, 39% cutting agencies. Chart: channel revenue time series.	Set the scene. Show the current state
2	The Gap	Divergence chart: 49% efficiency vs. 27% outcomes = 22-point gap.	Introduce the first problem. Efficiency does not equal results
3	The Disruption	Stacked area chart: AI Overview intent migration over 11 months.	Introduce the second problem. Search channels are being disrupted
4	Who Is Exposed	Industry bar chart + channel heatmap.	Show that the disruption is uneven, some channels are insulated
5	The Framework	Quadrant scatter plot + NAPS ranking under baseline.	Combine all three forces into a single decision view
6	Test Your Assumptions	Interactive scenario engine with parameter sliders.	Let the audience explore, turn the report into a reusable tool

Technical Complexity

Data Preparation

The data preparation for this project is moderate to high. It involves four separate work streams:

4. IAB (Dataset 1): Extracted revenue tables from 5 annual PDF reports. Harmonized channel names across years (IAB reclassified 'Retail Media' and 'Audio' categories during this period). Built a unified 5-year panel CSV. Calculated CAGR, share shift, and MMI scores.
5. SEMrush (Dataset 2): Manually transcribed monthly data tables and industry breakdowns from the blog post. Cross-referenced chart visuals against reported text. Flagged interpolated vs. confirmed data points. Calculated ADEI scores.
6. Gartner + CMO Survey (Dataset 3): Extracted metrics from press releases and PDF reports. Merged with FirstPageSage CAC benchmarks. Standardized industry and channel categories across sources.
7. Scenario Model (Dataset 4): Wrote a Python script (146 lines) to combine the three indices, cross them with industry/size/scenario dimensions, compute NAPS scores, and classify each combination. Validated output against CAC benchmarks.

Calculated Fields

We will use several calculated fields in Tableau. The most important ones:

- Market Momentum Index (MMI): Composite of 5-year CAGR (60% weight) and share trajectory (40% weight), normalized to 0–100. Already computed in the Python script and available as a column, but we will also recreate it as a Tableau calculated field so it updates with filters.
- Net Allocation Pressure Score (NAPS): ([MMI] + [AELI]) - [ADEI]. This is the core metric.
- Allocation Classification: IF [NAPS] >= 75 THEN 'Scale' ELSEIF [NAPS] >= 55 THEN 'Test' ELSEIF [NAPS] >= 35 THEN 'Hold' ELSE 'Reduce' END
- Efficiency-to-Outcome Gap: [Time_Efficiency_Pct] - [Business_Outcome_Pct]

Level of Detail (LOD) Calculations

- {FIXED [Channel] : AVG([MMI_Score])} – keeps the market momentum score stable for each channel regardless of what industry or scenario filters the user applies.
- {FIXED [Industry] : AVG([AELI_Score])} – enables industry-level comparison even when filtered to a single channel.
- {FIXED [Channel], [Scenario] : AVG([NAPS_Score])} – pre-computes the allocation score for each channel-scenario pair for smooth transitions when parameters change.

Parameters

Three Tableau parameters will power the scenario engine in Dashboard 3:

- AI Disruption Level (slider: 0.5x to 2.5x): multiplies the ADEI score to simulate scenarios where AI Overviews expand or contract.
- AI Efficiency Level (slider: 0.5x to 1.5x): multiplies the AELI score to simulate stronger or weaker AI productivity gains.
- Market Growth (slider: -15% to +15%): adjusts MMI to simulate bull or bear market conditions.

Sets and Context Filters

- Channel Health Set: classifies channels as 'Momentum-Positive' or 'Momentum-Negative' based on their IAB trajectory. Used to highlight channels that may score well on NAPS but have weakening fundamentals.
- High-Exposure Set: flags channels with ADEI above the 75th percentile.
- Context Filter on Industry: a global filter applied across all dashboards so the user can view everything through the lens of their specific industry.

Novel Chart Types

- Quadrant scatter plot with dynamic classification boundaries, the quadrant lines shift when scenario parameters change, and bubble colors update in real time. Reference: Few, S. (2012). Show Me the Numbers. Analytics Press.
- Divergence bar chart (butterfly chart), bars extend in opposite directions from a center axis. Reference: Knaflic, C. N. (2015). Storytelling with Data. Wiley.
- Stacked area chart for intent composition, proportional stacking to show compositional shift over time. Reference: Tufte, E. (2001). The Visual Display of Quantitative Information. Graphics Press.